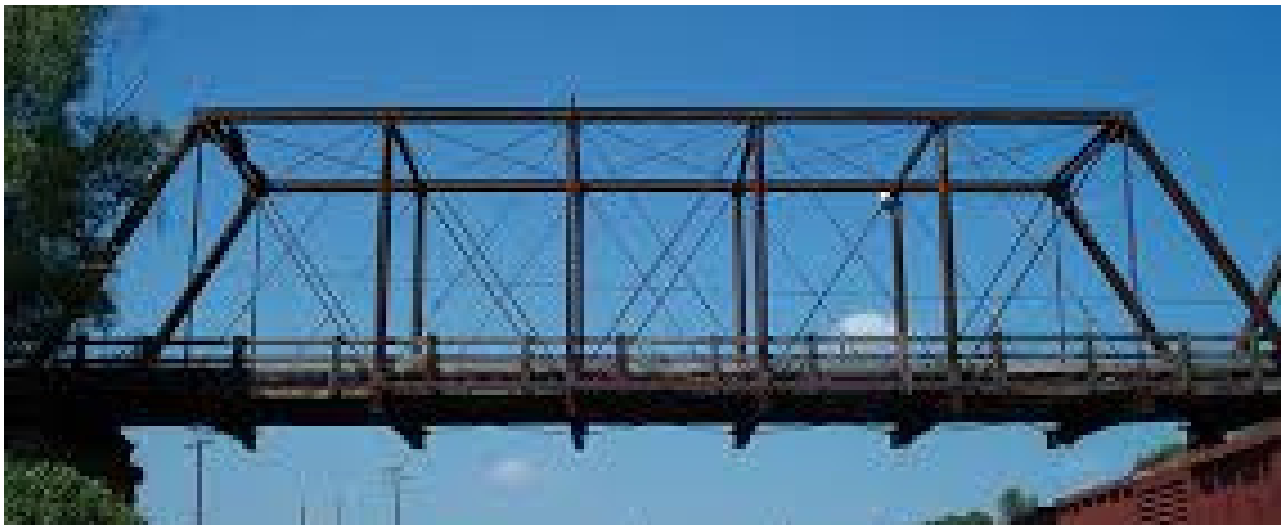


Indian Institute of Technology Patna



Group 02 Session: 2025 - 2026 Steel Structure Analysis Report



PRATT TRUSS

Course: CE3201 – Design of Steel Structures
Civil Engineering Dept.

PROJECT TITLE : PRATT TRUSS ANALYSIS

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OBJECTIVE

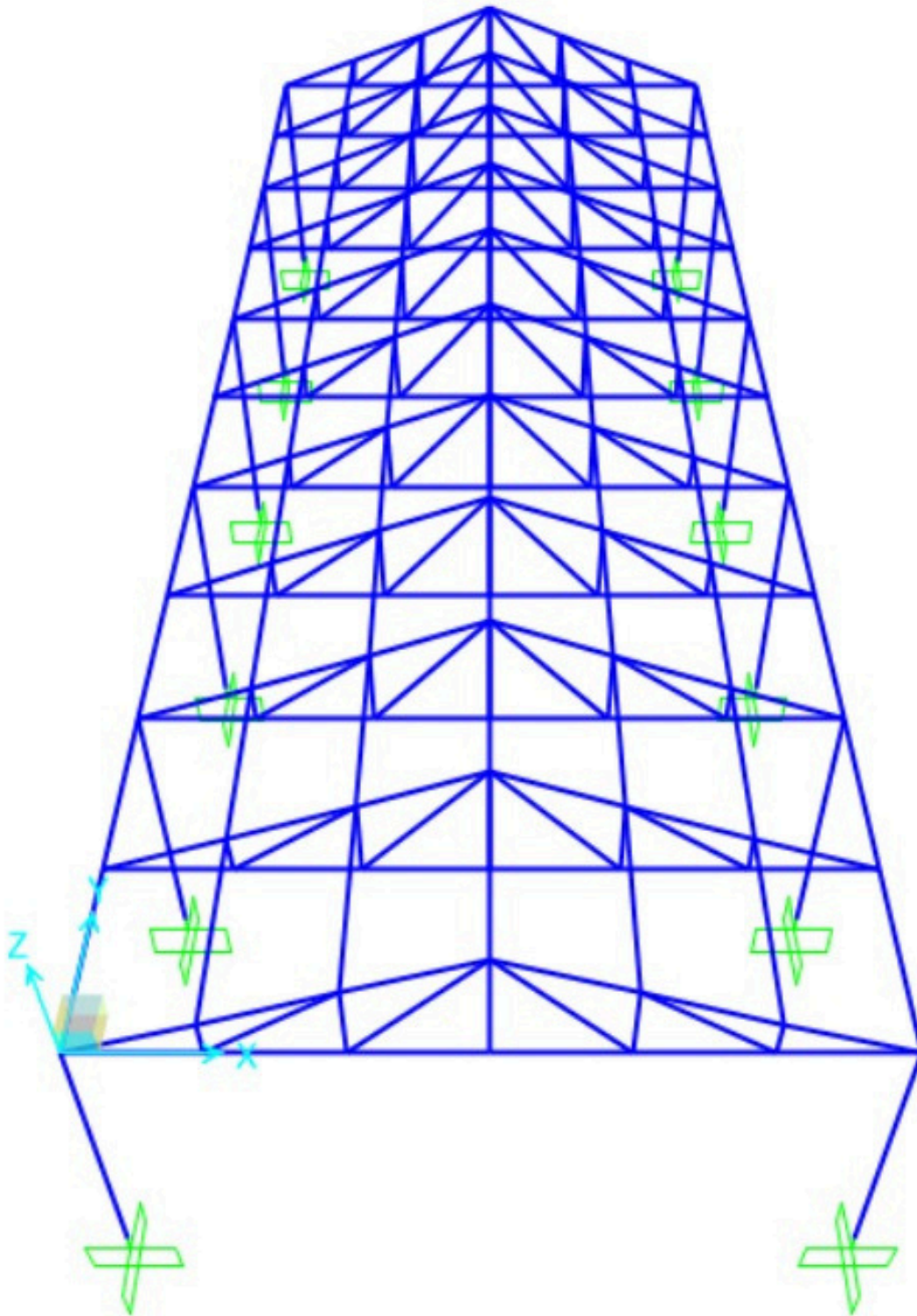
Designing an industrial building roof truss with the following features:

- Proposed location of the industrial building to be designed for 50 years of life – Agartala.
- Dimension of industrial shed – 50 m × 20 m
- Ceiling Height – 10 m (floor to roof truss bottom)
- Spacing of columns – 10 m (centre to centre)
- Spacing of trusses – not more than 5 m
- Rise of the roof truss – approx 4 m
- Truss type – Pratt Truss
- Column Height – 10 m
- Yield strength of steel members = 250 MPa
- Wind speed – 55 m/s (Agartala falls in terrain category 1)
- Wind Pressure – 1.905 kN/m²

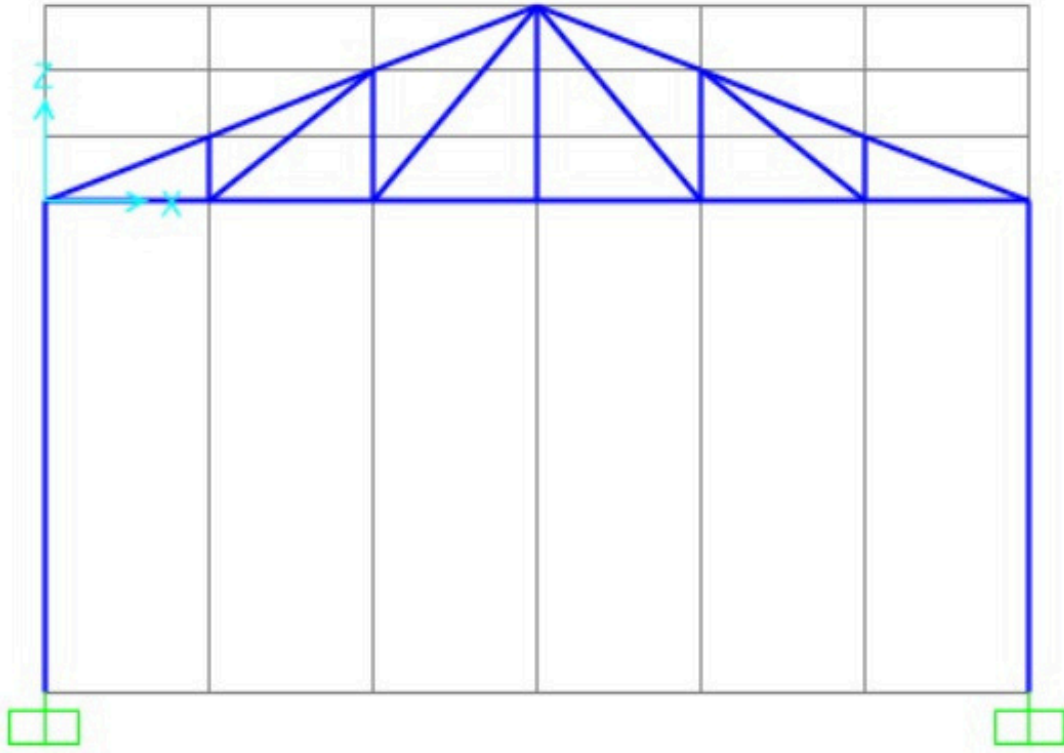
Load cases:

- Dead load – 1.5 kPa
- Live load – 1 kPa

3D View of Pratt Truss Geometry



2D View of Pratt Truss Geometry



IS: 875(Part 3): Wind Loads on Buildings and Structures – Proposed Draft & Commentary

WIND LOAD CALCULATION

5.3 – Design Wind Speed (v_z) = $v_b k_1 k_2 k_3 k_4$,

Where,

Basic wind speed, v_b = 55m/s

V_z = design wind speed at any height z in m/s,

k_1 = 1, probability factor (risk coefficient) (5.3.1) (Mean Probable design life of structure in years)

k_2 = 1.05, terrain roughness and height factor, Terrain category 1 (5.3.2)

k_3 = 1, topography factor (5.3.3)

k_4 = 1.15, importance factor for the cyclonic region for industrial strs. (5.3.4).

$$v_z = 1 \times 1.05 \times 1 \times 1.15 \times 55 = 66.4125 \text{ m/s}$$

5.4 – Design Wind Pressure

$$P_z = 0.6(v_z)^2 = 0.6 \times (66.4125)^2 = 2.646 \text{ kN/m}^2$$

where,

P_z = wind pressure in N/m² at height z, and

v_z = design wind speed in m/s at height z.

The design wind pressure P_d can be obtained as,

$$P_d = K_d \cdot K_a \cdot K_c \cdot P_z = 1 \times 0.9 \times 0.8 \times 2.646 = 1.905 \text{ kN/m}^2$$

Where,

K_d = 0.9 (Wind directionality factor)

K_a = 0.8 (Area averaging factor)

K_c = 1 (Combination factor)

(6.2.3.13)

Wind force (F):

$$F = (C_{pe} - C_{pi}) \times A_e \times P_d$$

$$h/w \leq 0.5$$

$$\alpha = \tan^{-1}(4/10)$$

$$\alpha = 21.80^\circ$$

For C_{pe} :

FOR 0°		FOR 90°	
EF	GH	EF	GH
- 0.33	-0.4	-0.7	-0.6

For C_{Pi} :

FOR 0°	
EF	GH
2	-2

So, $(C_{Pe} - C_{Pi})$:

FOR 0°	
EF	GH
- 0.9	-0.5

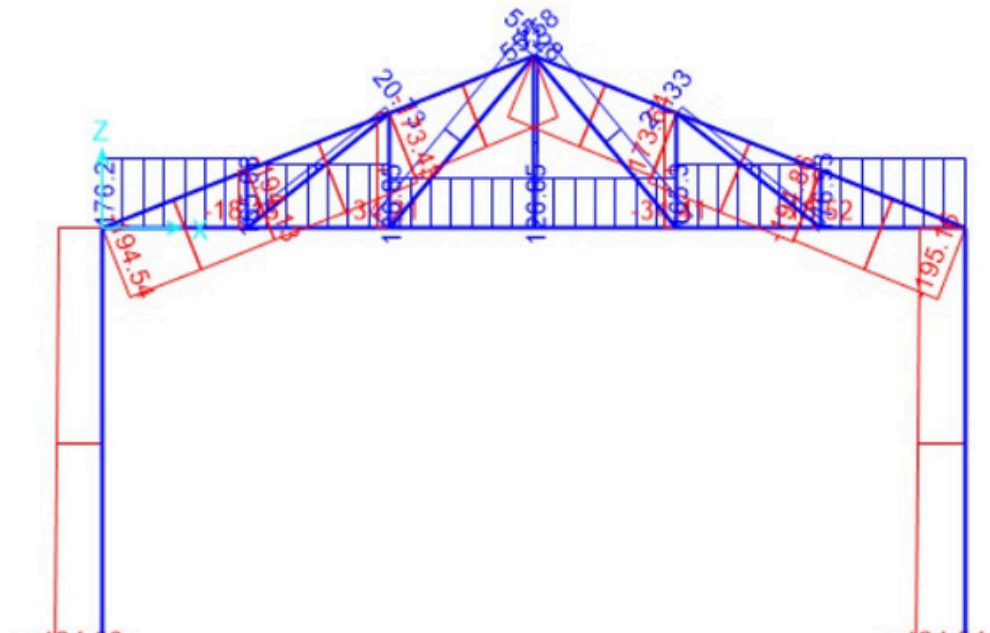
Perpendicular Distance between two purlin =

F_w = Force in the Windward direction per unit
length

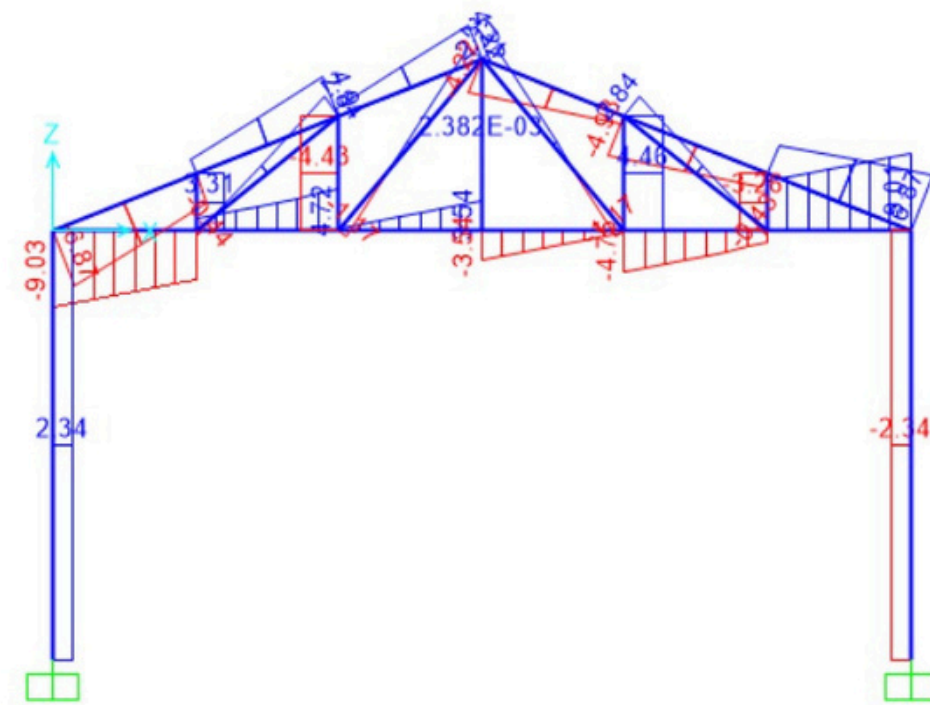
F_w = Force in the Leeward direction per unit
length

Load Case = $1.5(LL+DL)$

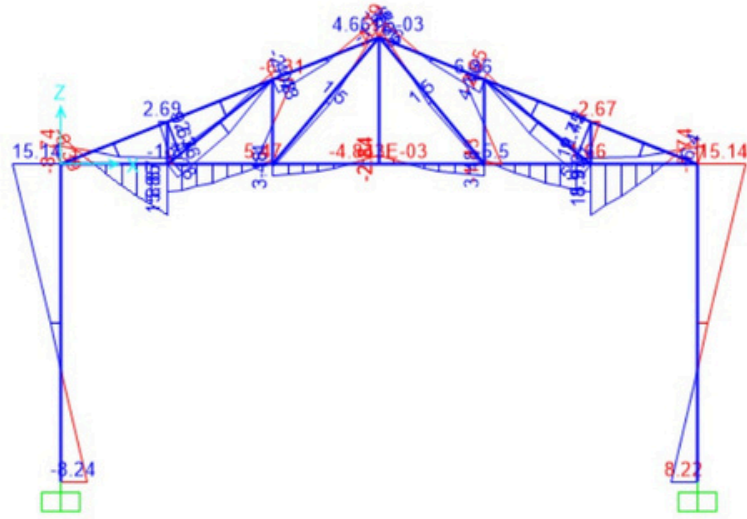
AXIAL FORCE



SHEAR FORCE

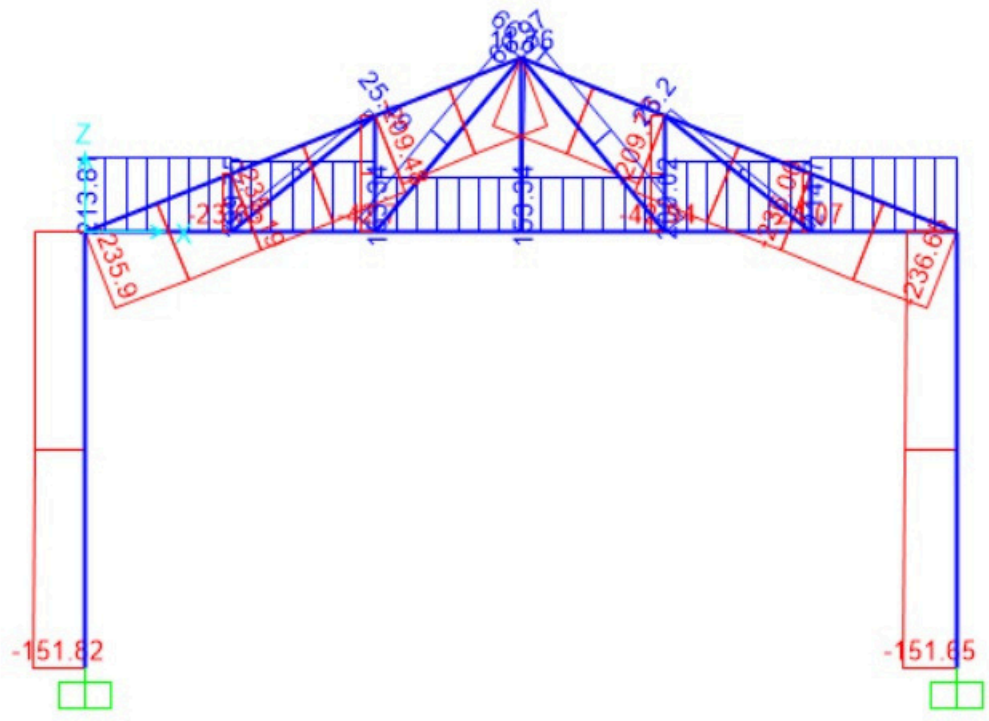


BENDING MOMENT

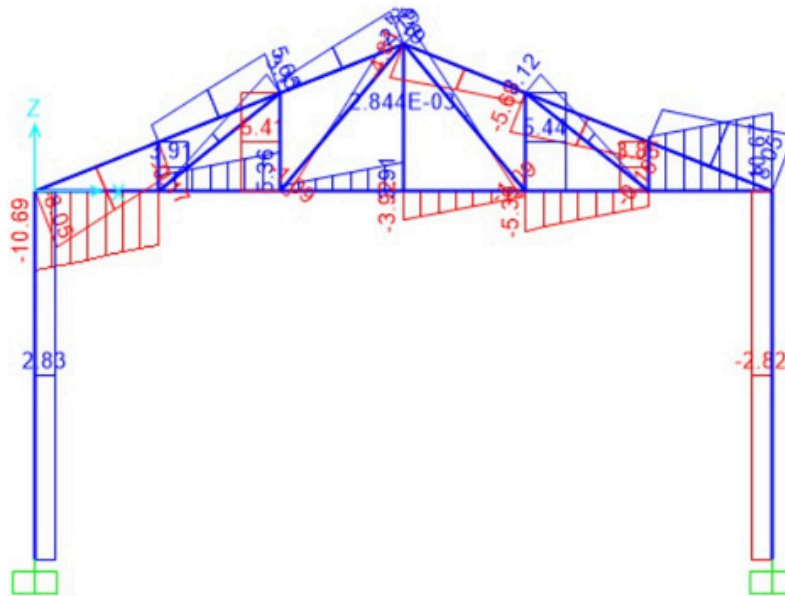


Load Case = $1.5(WL+DL)$

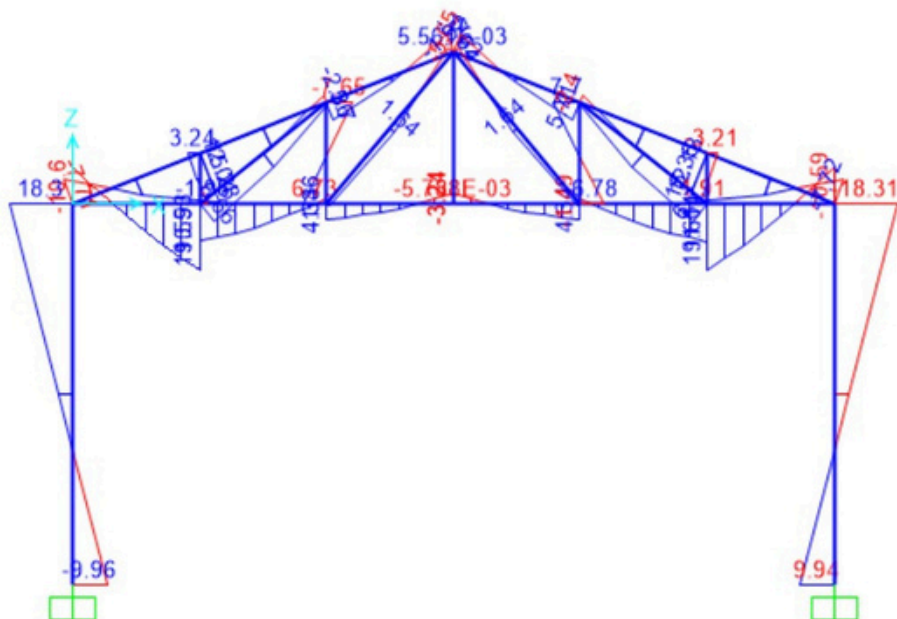
AXIAL FORCE



SHEAR FORCE



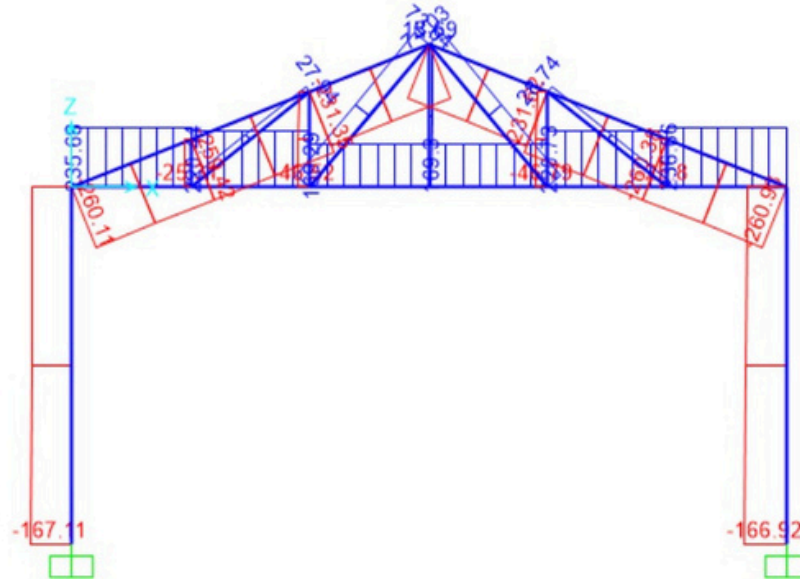
BENDING MOMENT



Load Case = $1.2(LL+DL+WL)$

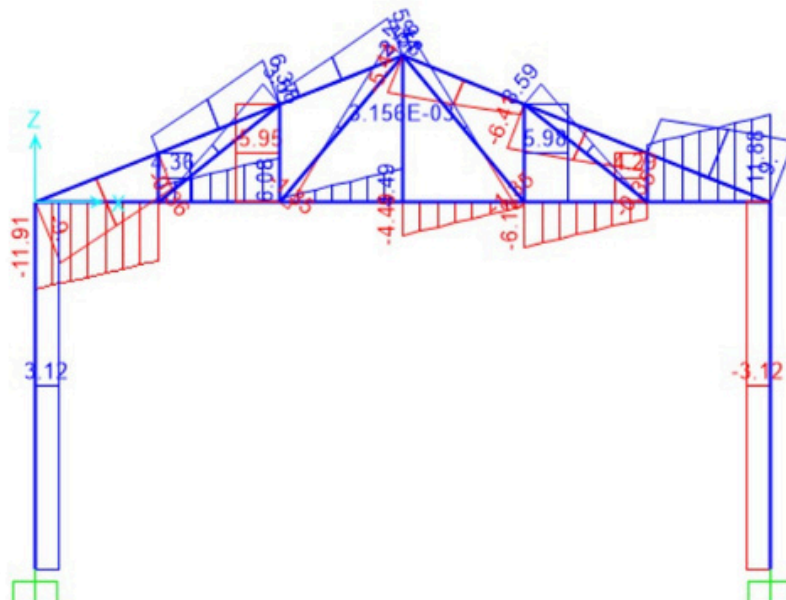
Axial Force Diagram (1.2(DL+LL+WL))

AXIAL FORCE



Shear Force 2-2 Diagram (1.2(DL+LL+WL))

SHEAR FORCE



BENDING MOMENT

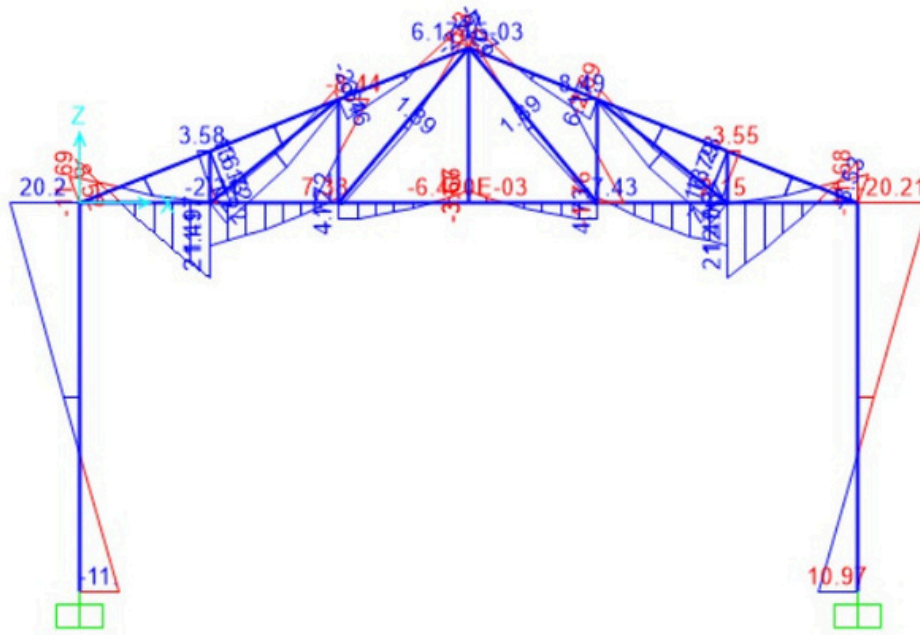


Table Forces in Critical Members

Member No.	Member	Load Type	Maximum Load (KN)	Load Case
1.	Top Chord	Axial Force	-1098.359	1.5(DL+LL)
			-1510.437	1.5(DL+WL)
			-1574.383	1.2(DL+LL+WL)
2.	Vertical	Axial Force	-222.5	1.5(DL+LL)
			-316.911	1.5(DL+WL)
			-327.306	1.2(DL+LL+WL)
3.	Diagonal	Axial Force	306.98	1.5(DL+LL)
			432.23	1.5(DL+WL)
			448.232	1.2(DL+LL+WL)
4.	Beam	Shear Force	-100.206	1.5(DL+LL)
			-130.913	1.5(DL+WL)
			-137.714	1.2(DL+LL+WL)
		Bending Moment	285.7508(kn-m)	1.5(DL+LL)
			406.2704(kn-m)	1.5(DL+WL)
			420.0579(kn-m)	1.2(DL+LL+WL)
5.	Purlin (Major)	Shear Force	-87.998	1.5(DL+LL)
			-125.545	1.5(DL+WL)
			-129.342	1.2(DL+LL+WL)
		Bending Moment	172.8284(kn-m)	1.5(DL+LL)
			-241.552(kn-m)	1.5(DL+WL)
			-250.3477(kn-m)	1.2(DL+LL+WL)
6.	Purlin (Minor)	Shear Force	-0.108	1.5(DL+LL)
			-11.545	1.5(DL+WL)
			-9.272	1.2(DL+LL+WL)
		Bending Moment	-0.3288(kn-m)	1.5(DL+LL)
			-9.946(kn-m)	1.5(DL+WL)
			-8.0663(kn-m)	1.2(DL+LL+WL)
7.	Horizontal	Axial Force	994.372	1.5(DL+LL)
			1356.182	1.5(DL+WL)
			1416.29	1.2(DL+LL+WL)
8.	Column	Axial Force	-769.066	1.5(DL+LL)
			-1040.95	1.5(DL+WL)
			-1088.91	1.2(DL+LL+WL)